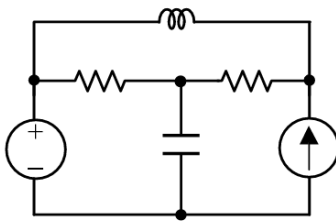
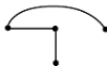
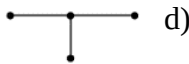
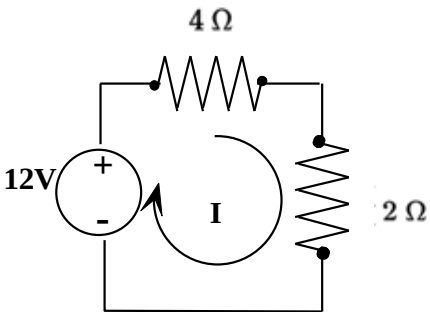


1.	Which of the following elements of electrical engineering cannot be analyzed using Ohm's law? a) Capacitors b) Inductors c) Transistors d) Resistance	c
2.	Which of the following according to KVL must be zero? a) Algebraic sum of currents in closed-loop b) Algebraic sum of power in closed-loop c) Algebraic sum of losses in closed-loop d) Algebraic sum of voltages in closed-loop	d
3.	Kirchhoff's Law is applicable to a) Passive network only b) A.C. Network only c) D.C. network only d) Both A.C and D.C circuits	d
4.	KCL works on the principle of which of the following a) law of conservation of charge b) Law of conservation of energy c) Both d) None of the above	a
5.	KVL works on the principle of a) Law of conservation of charge b) Law of conservation of energy c) Both d) None of the above	b
6.	Potential difference between two points of a wire carrying 2A current is 0.1V. The resistance of the wire is _____ a) 0.5 ohm b) 0.05 ohm c) 0.25 ohm d) 5 ohm	b
7.	Which of the following statements are true from the statements given below Statement 1: A tree is a graph but a graph is not a tree Statement 2: Both trees and graphs are non linear data structures a) Only statement 1 is true b) Only statement 2 is true c) Both statement 1 and statement 2 are true d) None of them is true	c

8.	The number of links for a graph having 'n' nodes and 'b' branches are a) $b-n+1$ b) $n-b+1$ c) $b+n-1$ d) $b+n$	a
9.	The tree branches of a graph are called a) Chords b) Links c) Twigs d) Co-tree	c
10.	The tie-set matrix gives the relation between a) Branch currents and link currents b) Branch voltages and link currents c) Branch currents and link voltages d) None of these	a
11.	Which one of the following statements is correct: A tree in a network is a connected graph containing a) All the nodes only b) All the branches only c) All the branches and nodes d) All the nodes but no close path	d
12.	The number of chords in the graph of the given circuit will be  a) 2 b) 4 c) 5 d) 6	a
13.	Consider the network graph shown in Figure, which one of the following is not a tree of this graph. a)  b)  c)  d) 	b

14.	What is a "branch" in the context of a network's graph representation? a) The complete circuit diagram b) A single closed loop c) A line segment connecting two nodes, representing a circuit element d) A matrix that defines the network	c
15.	15. In an incidence matrix, what do the rows and columns typically represent? a) Rows represent branches; columns represent nodes b) Rows represent loops; columns represent nodes c) Rows represent nodes; columns represent branches d) Rows represent loops; columns represent branches	c
16.	16. A graph without any cycles or closed loops is called as a) Complete graph b) Directed graph c) Planar graph d) Tree	d
17.	In a complete incidence matrix, the algebraic sum of the entries in any column is a) 0 b) 1 c) -1 d) 2	a
18.	A reduced incidence matrix is formed by removing one row from the complete incidence matrix. This removed row corresponds to the: a) Reference node b) Loop branch c) Independent loop d) Cut-set	a
19.	The rows of a tie-set matrix represent a) Node voltages b) Branch currents c) Tie-set currents (or link currents) d) Branch voltages	c
20.	For a connected graph with 'n' nodes, the number of independent cut-sets is a) n-1 b) n-b+1 c) b+n-1 d) b+n	a.
21.	Which of the following is a valid assumption in mesh analysis with only independent sources? a) Each node has the same voltage b) Each mesh must contain a voltage source c) Mesh currents can be assumed clockwise or counterclockwise d) Only KCL is needed	c
22.	In a DC circuit with only independent voltage sources and resistors, mesh analysis applies KVL around each loop. What is the correct KVL form? a) Sum of voltage drops across resistance and emf = zero b) Sum of currents at a junction = zero c) Sum of voltage drops across resistance = zero	a

	d) Current entering a node = current leaving	
23.	In nodal analysis of a circuit with only independent current sources & resistors, which of the following is correct? a) Current source is replaced with voltage b) Nodal voltages are solved using KCL c) Supernode is mandatory d) Voltage drops are summed in loops	b.
24.	A planar circuit has 4 meshes and contains only resistors and independent voltage sources. How many mesh equations are needed? a) 1 b) 3 c) 4 d) 5	c
25.	In a circuit with 3 meshes and 2 independent voltage sources, the number of mesh current equations is: a) 1 b) 2 c) 3 d) 5	c
26.	In a mesh with a voltage source of 12 V and resistors of $4\ \Omega$ and $2\ \Omega$ shown in Figure, what is the mesh current assuming clockwise direction?  a) 2 A b) 3 A c) 4 A d) 6 A	a
27.	Which of the following is not an expression power? a) $P=VI$ b) $P=I^2R$ c) $P=V^2/R$ d) $P=I/R$	d
28.	Number of twigs in a graph is calculated by a) $n-1$ b) $n-b+1$ c) $b+n-1$ d) $b+n$	a

29.	If a circuit has 'N' nodes, the number of independent node equations needed in nodal analysis is: a)N b)N-1 c)N+1 d)N-2	b
-----	---	---

Unit-2

1.	Rms value is defined based on which of the following a) Heating effect b) Charge transfer c) Current b) Voltage	a
2.	Which of the following defined the average value a) Voltage b) Heating effect c) Current d) Charge transfer	d
3.	For symmetrical waveform average value of one full cycle is a) 1 b) 1.11 c) 2.22 d) 0	d
4.	Inductor does not allow the sudden change of a) Current b) Voltage c) Power d) None of the above	a
5.	Capacitor does not allow the sudden change of a) Current b) Voltage c) Power d) None of the above	b
6.	If a pure inductor is connected across the ac source, the average power taken by the inductor is a) A few watt b) 100 watt c) Zero watt d) Maximum power	c
7.	Average power taken by the pure capacitor is a) Zero b) Minimum c) Maximum d) Any of the above	a
8.	In a series R, L circuit, voltage across resistor and inductor are 3 V and 4 V respectively, then what is the applied voltage a) 7V b) 5V c) 4V d) 3V	b
9.	The alternative names for active power is/are a) Real power b) Average power c) True power d) All of the above	d

10 .	In series R, L circuit power factor can be defined as a) R/Z b) P/S c) V_r/V d) All of the above	d
11 .	In ac RC series circuit total voltage is 10V and voltage across resistor is 6V, then what is voltage across capacitor a) 4V b) 8V c) 16V d) 10V	b
12 .	In series RLC circuit, voltage across resistor, inductor and capacitor are 5V, 2V and 2V respectively. Find total voltage a) 9V b) 4V c) 2V d) 5V	d
13 .	In RLC series circuit, if the voltage across the capacitor is greater than the voltage across inductor, then power factor of the network is a) Lagging b) Leading c) Unity d) Zero	b
14 .	An inductor works by a) Charging a piece of wire b) Storing Energy as a magnetic field c) Choking off DC d. Introducing Resistance into the field	b
15 .	The form factor is the ratio of a) Peak value to r.m.s. value b) r.m.s. value to average value c) Average value to r.m.s. value d) None of the above	b
16 .	Which of the following circuit components opposes the change in the circuit voltage a) Inductance b) Capacitance c) Conductance d) Resistance	b
17 .	For sine wave the value of form factor is a) 1 b) 1.11 c) 0 d) 1.414	b
18 .	For sine wave the value of peak factor is a) 1 b) 1.11	d

	c)0 d)1.414	
19 .	When a pure resistive circuit is energized by an AC supply, the angle between the voltage and its current is a)90° b) 0° c) In between 0° and 90° d) None of these	b
20 .	Power factor of a pure resistive circuit is a)1 b)0 c) in between 0 and 1 d)more than 1	a
21 .	In a pure inductive circuit, the ratio of voltage to current (V/I) is referred to as a) Resistance b) Admittance c) Inductive Reactance d) None of these	c
22 .	Power factor of a pure inductive circuit is a)1 b)0 c) in between 0 and 1 d)more than 1	b
23 .	The Inductive reactance is measured in a) henry b) farad c) ohm d) none of these	c
24 .	Power factor of a pure capacitive circuit is a)1 b)0 c) in between 0 and 1 d)more than 1	b
25 .	In a pure capacitive circuit, the ratio of voltage to current (V/I) is referred to as a)Resistance b) Admittance c)Capacitive Reactance d)None of these	c
26 .	The voltage induced in an inductor is represented as a) Product of its inductance and current through it b) Ratio of its inductance to current through it c) Ratio of current through it to its inductance d) Product of its inductance and rate of change of current through it	d
27 .	Power factor of a pure capacitive circuit is a. 0 b. 1	a

	c. in between 0 and 1 d. more than 1	
28	Kirchhoff's laws are useful in determining a. Current flowing in a circuit b. EMFs and Voltage drops in a circuit c. Power in a circuit d. All the above	d
29	In a pure capacitive circuit, the ratio of voltage to current (V/I) is referred to as a. Resistance b. Admittance c. Capacitive Reactance d. None of these	c
30	SI unit of power is a. ampere b. watt c. kilowatt-hour d. volt	b
31	Power factor of a pure resistive circuit is a. 1 b. 0 c. in between 0 and 1 d. more than 1	a
32	The power consumed by a pure resistor is a. Reactive Power b. Active Power c. Apparent Power d. None of these	b
33	Resistance/Impedance is equal to a. Form factor b. Peak factor c. Power factor d. None of these	c
34	In a pure inductive circuit, the ratio of voltage to current (V/I) is referred to as a. Resistance b. Admittance c. Inductive Reactance d. None of these	c
35	The Inductive reactance is measured in a. henry b. farad c. ohm d. None of these	c
36	When a pure resistive circuit is energized by an AC supply, the angle between the voltage and its current is a. 90°	b

	b. 0° c. In between 0 and 90° d. None of these	
37 .	Peak factor of a sinusoidal current is a. 0.707 b. 1.414 c. 1.11 d. None of these	b
38 .	According to KCL as applied to junction in a network a) Total sum of currents meeting at the junction is zero b) No current can leave the junction without some current entering it c) Net current flow at the junction is positive d) Algebraic sum of current meeting at the junction is zero	d
39 .	The power factor of an AC circuit is given by a) Cosine of the phase angle b) Tangent of the phase angle c) The ratio of R/XL d) The ratio of XL/Z	a
40 .	Kirchhoff Voltage law is concerned with a) Resistive drop b) Battery EMF c) Junction voltages d) Both a and b	d

Unit-3 4 and 5

Sl.No	MCQ Questions	Answer
1.	Which of the following is a non-conventional energy source? a. Coal b. Natural Gas c. Solar Energy d. Hydroelectric Power	c
2.	Which voltage level is commonly used for single-phase residential supply in India? a. 415 V b. 230 V c. 11 kV d. 132 kV	b
3.	The device which converts solar energy directly into electricity is a. Solar heater b. Photovoltaic cell c. Solar concentrator d. Solar chimney	b
4.	What voltage level is considered as extra high voltage in transmission systems? a. 33 kV – 66 kV b. 66 kV – 132 kV c. 220 kV – 400 kV d. Above 400 kV	d
5.	In a thermal power plant, the primary energy conversion is from: a. Mechanical to electrical b. Chemical to thermal to electrical c. Nuclear to thermal to electrical d. Kinetic to electrical	b
6.	In a thermal power plant, the boiler converts a. Electrical energy to thermal energy b. Thermal energy to mechanical energy c. Chemical energy to thermal energy d. Mechanical energy to electrical energy	c
7.	In a nuclear power plant, the main function of the moderator is to a. Absorb neutrons b. Slow down fast neutrons c. Cool the reactor d. Increase neutron speed	b
8.	The main disadvantage of wind energy is a. High operating cost	c

	b. Pollution c. Intermittent and unpredictable availability d. Complex generator design	
9.	What is the function of a generator in a power plant? a. To step up the voltage for transmission. b. To convert mechanical energy into electrical energy. c. To burn fuel and produce heat. d. To cool down the steam from the turbine.	b
10.	In a hydroelectric power plant, the potential energy of water is first converted into a. Heat energy b. Kinetic energy c. Chemical energy d. electrical energy directly	b.
11.	The voltage generated at most power plants is relatively low and is stepped up using which of the following devices for efficient transmission? a. Capacitor b. Generator c. Transformer d. Turbine	c
12.	13. What is a key advantage of HVAC transmission over HVDC? a. Lower power loss b. Terminal equipment is less expensive c. It can be used for interconnecting different frequency systems d. No reactive power is consumed	b
13.	The secondary distribution system is primarily responsible for supplying electricity to: (a) Large industrial consumers (b) The main grid substation (c) Small industrial, commercial, and residential consumers (d) Other distribution substations	c
14.	HVAC transmission systems typically use how many conductors? a) One b) Two c) Three d) Four	c
15.	Which of these is NOT a basic component of a thermal power plant? a) Boiler b) Turbine c) Cooling tower d) Control room	d
16.	The voltage range for Low Voltage (LV) is typically considered to be a) Below 1000 V b) 1 kV to 100 kV c) Above 100 kV d) Greater than 765 kV	a
17.	The primary purpose of the step-up transformer in a power plant is to: a) Increase the voltage for transmission	a

	b) Decrease the voltage for distribution c) Generate electricity d) Condense steam back into water	
18	The process of stepping down the voltage from the primary distribution level (e.g., 11 kV) to the secondary distribution level (e.g., 415 V) is carried out by what component? a) Step-up transformer b) Distribution transformer c) Voltage regulator d) Isolator	b
19.	Which type of load is typically supplied by a 415 V, three-phase power system? a) Domestic lighting b) Small industrial and commercial loads c) Residential appliances d) Streetlights	b
20.	What does SMPS stand for? A) Switched Mode Power System B) Switched Mode Power Supply C) Smart Mode Power Solution D) Simple Mode Power Source	b
21	What is the primary function of an SMPS? A) To convert AC to DC power B) To convert DC to AC power C) To regulate power supply D) To generate power	a
22.	What does PMSM stand for? A) Permanent Magnet Synchronous Motor B) Permanent Magnet Stepper Motor C) Pulse Width Modulation Switching Motor D) Power Management Switching Module	a
23.	What is a significant cost factor that makes EVs more economical in the long run? a) Higher battery replacement costs b) Lower operating costs (fuel and maintenance) c) Higher insurance costs d) Lower resale value	b
24.	What is the primary characteristic of a PMSM? A) Uses electromagnetic induction B) Has a permanent magnet rotor C) Requires a commutator D) Uses a brushed motor design	b
25.	What is the advantage of PMSM over induction motors? A) Higher efficiency B) Lower cost C) Simpler design D) Larger size	a
26.	What is the typical application of PMSM?	c

	A) Household appliances B) Industrial power systems C) Electric vehicles and robotics D) Aerospace and defense	
27.	What is the key benefit of PMSM in electric vehicles? A) Higher torque density B) Lower noise emission C) Improved fuel efficiency D) Reduced maintenance	a
28.	Two main parts of a PMSM are:- A) Field, Armature B) Stator, Armature C) Stator, Rotor D) Field, Rotor	c
29.	Two types of BLDC motors are :- A) Slip ring BLDC Motor, Squirrel cage BLDC Motor B) Series BLDC Motor, Shunt BLDC Motor C) Unipolar BLDC Motor, Bipolar BLDC Motor D) Short shunt BLDC Motor, Long shunt BLDC Motor	c
30.	Boost converter is a _____ converter A) Step down - Step up B) Step down C) Step Up - Step down D) Step up	d
31.	Boost converter uses _____ A) An inductor & a diode only B) A capacitor & a diode only C) An inductor, a capacitor & a diode D) An inductor & a capacitor only	c
32.	In a boost converter, output voltage is _____ A) Input Voltage / (1-D) B) $D \times \text{Input Voltage}$ C) $(1-D) / \text{Input Voltage}$ D) $(1-D) \times \text{Input Voltage}$	a
33.	Buck converter is also called as _____ A) Step up converter B) Step down converter C) Step Up - Step down converter D) Step down - step up converter	b
34.	Output voltage of a step down converter is ____ A) Input Voltage / D B) $(1-D) \times \text{Input Voltage}$ C) $D \times \text{Input Voltage}$ D) Input Voltage / (1-D)	c
35.	13. Which component is the primary source of energy storage in an electric vehicle a) Electric Motor	b

	b) Battery Pack c) Power Electronics Controller d) Charging Port	
36.	The component responsible for converting the DC power from the battery to AC power for the electric motor is the: a) Onboard Charger b) DC-DC Converter c) Power Electronics Controller (Inverter) d) Thermal System	c
37.	What is the primary function of the electric motor in an EV? a) To store electrical energy b) To convert electrical energy into mechanical energy to propel the vehicle c) To convert AC power from the grid to DC power for the battery d) To manage the temperature of the battery pack	b
38.	Which component allows an EV to capture and store energy that is normally lost during braking? a) Regenerative Braking System b) DC-DC Converter c) Power Electronics Controller d) Thermal System	a
39.	What is the role of the charging port? a) To store energy for the vehicle b) To provide power to the electric motor c) To connect the vehicle to an external power source for recharging the battery d) To convert AC to DC power for the battery	c
40	What component in a BLDC motor replaces the mechanical commutator and brushes found in a conventional DC motor? a) A flywheel b) An electronic controller c) A gearbox d) A universal joint	b
41.	A Permanent Magnet Synchronous Motor (PMSM) is analogous in construction to a: a) Induction motor b) Conventional brushed DC motor c) Switched reluctance motor d) Brushless DC (BLDC) motor	d
42.	Which motor type is typically the most expensive due to the use of rare-earth magnets? a) Switched Reluctance Motor (SRM) b) Induction Motor (IM) c) Permanent Magnet Synchronous Motor (PMSM) d) DC Series Motor	c
43.	Faraday's First Law states: (a) The changing magnetic field linked with a conductor induces an electromotive force in the conductor.	a

	(b) The induced emf is inversely proportional to the rate of change of magnetic flux linkage. (c) The induced emf is directly proportional to the magnetic flux linkage. (d) The induced emf is inversely proportional to the magnetic flux linkage.	
44.	Faraday's Second Law states: (a) The magnitude of induced emf in a closed circuit is directly proportional to the rate of change of magnetic flux linked with the circuit. (b) The direction of the induced emf is such that it aids the change that produced it. (c) The magnitude of the induced emf is independent of the change that produced it. (d) The direction of the induced emf is always clockwise.	a
45.	Self-induced emf is produced when: (a) The current in a coil changes. (b) The current in a nearby coil changes. (c) A magnet moves near a coil. (d) A coil moves in a magnetic field.	a
46.	Mutually induced emf is produced when: (a) The current in a coil changes. (b) The current in a nearby coil changes. (c) A magnet moves near a coil. (d) A coil moves in a magnetic field.	b
47.	The following are the parts of a DC motor are: (a) Stator, rotor, commutator, brushes (b) Stator, rotor, slip rings, brushes (c) Stator, rotor, commutator, slip rings (d) Stator, rotor, slip rings, commutator	a
48.	The principle of operation of a DC motor is based on: (a) Faraday's Law of Electromagnetic Induction (b) Lenz's Law (c) Ohm's Law (d) Kirchhoff's Law	a
49.	Which of the following is NOT a type of DC motor? (a) Shunt motor (b) Series motor (c) Compound motor (d) Synchronous motor	d
50.	Transformers are used to: (a) Change the voltage level of AC power (b) Change the frequency of AC power (c) Convert AC power to DC power (d) Convert DC power to AC power	a
51.	The main advantage of using transformers in power transmission is: (a) Reduced power losses (b) Increased power transmission efficiency (c) Ability to transmit power over long distances (d) All of the above	d

52.	The main advantage of using transformers in power transmission is: (a) Reduced power losses (b) Increased power transmission efficiency (c) Ability to transmit power over long distances (d) All of the above	c
53.	Which of the following is NOT a type of transformer? (a) Step-up transformer (b) Step-down transformer (c) Autotransformer (d) DC transformer	d
54.	A transformer has 500 turns in the primary winding and 100 turns in the secondary winding. If the primary voltage is 220V, the secondary voltage will be: (a) 44V (b) 1100V (c) 110V (d) 22V	a
55.	What is the fundamental principle behind a stepper motor's operation? a) Induction through AC current b) Electromagnetic attraction/repulsion causing discrete steps c) Faraday's law of induction d) Friction between rotor and stator	b
56.	What does MCB stand for? A) Miniature Circuit Breaker B) Main Control Board C) Motor Control Box D) Maximum Current Breaker	a
57.	What is the primary function of an MCB? A) To supply power to a circuit B) To control the speed of a motor C) To protect a circuit from overcurrent D) To regulate voltage levels	c
58.	Which of the following is a benefit of a proper earthing system? A) Reduced risk of fire B) Improved system efficiency C) Protection of personnel from electric shock D) Increased system capacity	c
59.	What happens when a leakage current flows to ground through an earthing system? A) The circuit breaker trips B) The fuse blows C) The voltage is reduced D) The current is safely dissipated	d
60.	A fuse is a: A) One-time use device that needs replacement after blowing B) Resettable device like a circuit breaker C) Mechanical switching device	a

	D) Device that only works on high voltage	
61	For plate earthing using a copper plate, the size of the plate should not be less than: a) 60 cm x 60 cm x 6.35 mm b) 60 cm x 60 cm x 3.18 mm c) 30 cm x 30 cm x 3.18 mm d) 60 mm x 60 mm x 3.18 mm	b
62.	If a transformer has a turns ratio of 2:1, it means: a) It is a step up transformer b) Voltage on secondary is double c) Voltage on primary is double d) Transformer is faulty	c

Sl. No	MCQ Questions	Answer
1	Rms value is defined based on which of the following a) Heating effect b) Charge transfer c) Current b) Voltage	a
2	Which of the following defined the average value a) Voltage b) Heating effect c) Current d) Charge transfer	d
3	For symmetrical waveform average value of one full cycle is a) 1 b) 1.11 c) 2.22 d) 0	d
4	Inductor does not allow the sudden change of a) Current b) Voltage c) Power d) None of the above	a
	Capacitor does not allow the sudden change of a) Current b) Voltage c) Power d) None of the above	b
6	If a pure inductor is connected across the ac source, the average power taken by the inductor is a) A few watt b) 100 watt c) Zero watt d) Maximum power	c
7	Average power taken by the pure capacitor is a) Zero b) Minimum c) Maximum	a

	d) Any of the above	
8	In a series R, L circuit, voltage across resistor and inductor are 3 V and 4 V respectively, then what is the applied voltage a) 7V b) 5V c) 4V d) 3V	b
9	The alternative names for active power is/are a) Real power b) Average power c) True power d) All of the above	d
10	In series R, L circuit power factor can be defined as a) R/Z b) P/S c) V_r/V d) All of the above	d
11	In ac RC series circuit total voltage is 10V and voltage across resistor is 6V, then what is voltage across capacitor a) 4V b) 8V c) 16V d) 10V	b
12	In series RLC circuit, voltage across resistor, inductor and capacitor are 5V, 2V and 2V respectively. Find total voltage a) 9V b) 4V c) 2V d) 5V	d
13	In RLC series circuit, if the voltage across the capacitor is greater than the voltage across inductor, then power factor of the network is a) Lagging b) Leading c) Unity d) Zero	b
14	An inductor works by a) Charging a piece of wire b) Storing Energy as a magnetic field c) Choking off DC d. Introducing Resistance into the field	b
15	The form factor is the ratio of	b

	a) Peak value to r.m.s. value b) r.m.s. value to average value c) Average value to r.m.s. value d) None of the above	
16	Which of the following circuit components opposes the change in the circuit voltage a) Inductance b) Capacitance c) Conductance d) Resistance	b
17	For sine wave the value of form factor is a) 1 b) 1.11 c) 0 d) 1.414	b
18	For sine wave the value of peak factor is a) 1 b) 1.11 c) 0 d) 1.414	d
19	When a pure resistive circuit is energized by an AC supply, the angle between the voltage and its current is a) 90° b) 0° c) In between 0° and 90° d) None of these	b
20	Power factor of a pure resistive circuit is a) 1 b) 0 c) in between 0 and 1 d) more than 1	a
21	In a pure inductive circuit, the ratio of voltage to current (V/I) is referred to as a) Resistance b) Admittance c) Inductive Reactance d) None of these	c
22	Power factor of a pure inductive circuit is a) 1 b) 0 c) in between 0 and 1 d) more than 1	b
23	The Inductive reactance is measured in a) henry b) farad c) ohm	c

	d) none of these	
24	Power factor of a pure capacitive circuit is a)1 b)0 c) in between 0 and 1 d)more than 1	b
25	In a pure capacitive circuit, the ratio of voltage to current (V/I) is referred to as a)Resistance b) Admittance c)Capacitive Reactance d)None of these	c
26	The voltage induced in an inductor is represented as a) Product of its inductance and current through it b) Ratio of its inductance to current through it c) Ratio of current through it to its inductance d) Product of its inductance and rate of change of current through it	d
27	Power factor of a pure capacitive circuit is a. 0 b. 1 c. in between 0 and 1 d. more than 1	a
28	Kirchhoff's laws are useful in determining a. Current flowing in a circuit b. EMFs and Voltage drops in a circuit c. Power in a circuit d. All the above	d
29	In a pure capacitive circuit, the ratio of voltage to current (V/I) is referred to as a. Resistance b. Admittance c. Capacitive Reactance d. None of these	c
30	SI unit of power is a. ampere b. watt c. kilowatt-hour d. volt	b
31	Power factor of a pure resistive circuit is a.1 b. 0 c. in between 0 and 1 d. more than 1	a
32	The power consumed by a pure resistor is a.Reactive Power b. Active Power c. Apparent Power	b

	d. None of these	
33	Resistance/Impedance is equal to a. Form factor b. Peak factor c. Power factor d. None of these	c
34	In a pure inductive circuit, the ratio of voltage to current (V/I) is referred to as a. Resistance b. Admittance c. Inductive Reactance d. None of these	c
35	The Inductive reactance is measured in a. henry b. farad c. ohm d. None of these	c
36	When a pure resistive circuit is energized by an AC supply, the angle between the voltage and its current is a. 90° b. 0° c. In between 0 and 90° d. None of these	b
37	Peak factor of a sinusoidal current is a. 0.707 b. 1.414 c. 1.11 d. None of these	b
38	The power factor of an AC circuit is given by a) Cosine of the phase angle b) Tangent of the phase angle c) The ratio of R/XL d) The ratio of XL/Z	a
39	Average value of fully rectified sine wave is a) 0 b) 0.637 times of its maximum value c) 0.318 times of its maximum value d) 0.5 times of its maximum value	b
40	Average value of half rectified sine wave is a) 0 b) 0.637 times of its maximum value c) 0.318 times of its maximum value d) 0.5 times of its maximum value	c

Sl. No	MCQ Questions	Answer
1.	Which of the following is a non-conventional energy source? a. Coal b. Natural Gas c. Solar Energy d. Hydroelectric Power	c
2.	Which voltage level is commonly used for single-phase residential supply in India? a. 415 V b. 230 V c. 11 kV d. 132 kV	b
3.	The device which converts solar energy directly into electricity is a. Solar heater b. Photovoltaic cell c. Solar concentrator d. Solar chimney	b
4.	What voltage level is considered as extra high voltage in transmission systems? a. 33 kV – 66 kV b. 66 kV – 132 kV c. 220 kV – 400 kV d. Above 400 kV	d
5.	In a thermal power plant, the primary energy conversion is from: a. Mechanical to electrical b. Chemical to thermal to electrical c. Nuclear to thermal to electrical d. Kinetic to electrical	b
6.	In a thermal power plant, the boiler converts a. Electrical energy to thermal energy b. Thermal energy to mechanical energy c. Chemical energy to thermal energy d. Mechanical energy to electrical energy	c
7.	In a nuclear power plant, the main function of the moderator is to a. Absorb neutrons b. Slow down fast neutrons c. Cool the reactor d. Increase neutron speed	b
8.	The main disadvantage of wind energy is a. High operating cost b. Pollution c. Intermittent and unpredictable availability	c

	d. Complex generator design	
9.	What is the function of a generator in a power plant? a. To step up the voltage for transmission. b. To convert mechanical energy into electrical energy. c. To burn fuel and produce heat. d. To cool down the steam from the turbine.	b
10.	In a hydroelectric power plant, the potential energy of water is first converted into a. Heat energy b. Kinetic energy c. Chemical energy d. electrical energy directly	b.
11.	The voltage generated at most power plants is relatively low and is stepped up using which of the following devices for efficient transmission? a. Capacitor b. Generator c. Transformer d. Turbine	c
12.	13. What is a key advantage of HVAC transmission over HVDC? a. Lower power loss b. Terminal equipment is less expensive c. It can be used for interconnecting different frequency systems d. No reactive power is consumed	b
13.	The secondary distribution system is primarily responsible for supplying electricity to: (a) Large industrial consumers (b) The main grid substation (c) Small industrial, commercial, and residential consumers (d) Other distribution substations	c
14.	HVAC transmission systems typically use how many conductors? a) One b) Two c) Three d) Four	c
15.	Which of these is NOT a basic component of a thermal power plant? a) Boiler b) Turbine c) Cooling tower d) Control room	d
16.	The voltage range for Low Voltage (LV) is typically considered to be a) Below 1000 V b) 1 kV to 100 kV c) Above 100 kV d) Greater than 765 kV	a
17.	The primary purpose of the step-up transformer in a power plant is to: a) Increase the voltage for transmission b) Decrease the voltage for distribution c) Generate electricity	a

	d) Condense steam back into water	
18.	The process of stepping down the voltage from the primary distribution level (e.g., 11 kV) to the secondary distribution level (e.g., 415 V) is carried out by what component? a) Step-up transformer b) Distribution transformer c) Voltage regulator d) Isolator	b
19.	Which type of load is typically supplied by a 415 V, three-phase power system? a) Domestic lighting b) Small industrial and commercial loads c) Residential appliances d) Streetlights	b
20.	What is the primary method used to generate electricity from hydro energy? a) By burning fuels to create steam that spins a generator. b) By capturing solar radiation and converting it into mechanical energy. c) By converting the mechanical energy of moving or falling water into electrical energy using turbines and generators. d) By extracting chemical energy from water molecules through a high-efficiency combustion process.	c
21.	Which component of a hydroelectric system is specifically designed to prevent debris from damaging the gates and turbine runners? a) Surge Tank b) Trash Rack c) Tailrace d) Forebay	b
22.	What is the primary function of a Surge Tank within the water-conduit system? a) To store water for irrigation purposes. b) To discharge water back into the river. c) To act as a pressure-release valve preventing the "water hammer" effect. d) To discharge excess reservoir water to prevent overtopping.	c
23.	Which part of the system is responsible for converting the kinetic energy of striking water into mechanical energy? a) The Generator b) The Turbine (Prime Mover) c) The Penstock d) The Spillway	b
24.	What is the purpose of the Spillway in the architecture of a dam? a) To carry water from the reservoir to the turbine. b) To store temporary water for generation. c) To act as a safety valve by discharging excess reservoir water to prevent overtopping. d) To house the control systems and generators.	c
25.	What two ways can nuclear energy be released from the nucleus of an atom? a) Through chemical combustion or thermal evaporation. b) By fission, where the nucleus splits into parts, or by fusion, where nuclei join together. c) By converting mechanical energy into kinetic energy. d) Through the oxidation of protons and neutrons.	b

26.	In a nuclear power plant, what is the primary function of the nuclear reactor? a) To cool the steam back into water for recirculation. b) To split uranium atoms through fission to release heat energy. c) To convert mechanical energy into electrical energy. d) To transfer heat directly to the turbine blades.	b
27.	Which component is responsible for transferring the heat from the reactor to water to create steam? a) The Condenser b) The Generator c) The Heat exchanger d) The Alternator	c
28.	Based on the mechanics of nuclear power generation, what is the specific role of the turbine? a) It cools the water to prevent overheating. b) It uses high-pressure steam to spin blades and generate mechanical energy. c) It subjects uranium to the fission process. d) It converts electrical energy back into mechanical energy.	b
29.	What is the purpose of the condenser within the nuclear power system? a) It produces electricity at its output. b) It cools the steam back into water so it can be recirculated. c) It synchronizes the turbine with the alternator. d) It increases the pressure of the steam before it hits the turbine	b
30.	How is electricity ultimately produced at the output of a nuclear power station? a) By the direct splitting of uranium atoms in the generator. b) By the heat exchanger transferring steam to the condenser. c) By a turbine synchronized with an alternator (synchronous generator). d) By the condenser releasing thermal energy into the atmosphere	c
31.	Which fuel is primarily used in thermal power plants a) Natural gas b) Coal c) Crude oil d) Nuclear uranium	b
32.	What is the correct sequence of energy conversion in a thermal power plant? a) Mechanical energy → Thermal energy → Chemical energy → Electrical energy b) Chemical energy → Heat energy → Thermal energy → Mechanical energy → Electrical energy c) Electrical energy → Mechanical energy → Chemical energy → Thermal energy d) Thermal energy → Chemical energy → Mechanical energy → Electrical energy	b
33.	What is the function of a Superheater in the steam generation process? a) It uses flue gases to heat combustion air. b) It converts the steam's thermal energy into mechanical energy. c) It takes steam from the boiler and further heats it to a high temperature to increase its energy content. d) It collects the leftover ash from coal combustion.	c
34.	How does an Air preheater contribute to the boiler's operation? a) It pumps water into the boiler tubes.	b

	b) It uses hot flue gases leaving the boiler to preheat combustion air before it enters the furnace. c) It extracts steam from the turbine to heat feed water. d) It stores coal before it is transported for combustion.	
35.	Which component is responsible for reheating steam after it has already passed through an initial stage of the turbine? a) Feed water heater b) Reheater c) Superheater d) Cooling Tower	b
36.	In a solar PV power plant, what is the primary mechanism that generates direct current (DC)? a) High-pressure steam spinning a turbine. b) Photons from sunlight striking semiconductor material and releasing electrons within solar cells. c) Heating a fluid like molten salt to produce a chemical reaction. d) Using heliostats to focus light onto a central generator.	b
37.	What is the specific role of an inverter within a solar energy system? a) To regulate the flow of electricity to prevent battery overcharging. b) To concentrate sunlight onto a single receiver point. c) To convert the direct current (DC) generated by panels into usable alternating current d) To cool steam back into water for recirculation.	c
38.	Which component of a solar thermal system is responsible for protecting a battery and extending its lifespan? a) The Controller, which regulates electricity flow to prevent overcharging and deep discharge. b) The Inverter, which switches energy types for consumption. c) The Heat-transfer fluid, which absorbs excess thermal energy. d) The Receiver, which absorbs concentrated sunlight.	a
39.	In a solar thermal plant, what are the two main types of solar collectors used to reflect and concentrate sunlight? a) Photovoltaic cells and semiconductor panels. b) Heliostats and parabolic troughs. c) Inverters and synchronous generators. d) Condensers and thermal tanks.	b
40	What happens to the steam in a solar thermal cycle after it has passed through the turbine? a) It is released into the atmosphere as thermal energy. b) It is cooled and condensed back into water, then pumped back to the steam generator to repeat the cycle. c) It is converted into direct current by the solar panels. d) It is stored in a thermal tank to be used as molten salt later.	b

UNIT-4

1. What is the primary function of a buck converter?

- A. Step up DC voltage
- B. Convert AC to DC
- C. Step down DC voltage**
- D. Convert DC to AC

2. In a buck converter, which component stores energy when the switch is ON?

- A. Diode
- B. Inductor**
- C. Capacitor
- D. Resistor

3. The output voltage of an ideal buck converter is primarily controlled by:

- A. Switching frequency
- B. Load resistance
- C. Duty cycle of the switch**
- D. Inductor value

4. Which of the following is a common application of a buck converter?

- A. Battery charging in solar systems
- B. Voltage regulation in mobile phone chargers**
- C. High-voltage transmission lines
- D. RF signal amplification

5. When the switch in a buck converter is turned OFF, the inductor current:

- A. Immediately drops to zero
- B. Flows through the diode to the load**
- C. Reverses direction
- D. Charges the switch

6. What is the main function of a boost converter?

- A. Step down DC voltage
- B. Step up DC voltage**
- C. Convert AC to DC
- D. Convert DC to AC

7. In a boost converter, energy is stored in which component when the switch is ON?

- A. Capacitor
- B. Diode

C. Inductor

D. Transformer

8. The output voltage of an ideal boost converter is controlled by:

A. Input voltage

B. Load resistance

C. Duty cycle of the switch

D. Output capacitor value

9. Which of the following is a common application of a boost converter?

A. Laptop power supply

B. LED driver circuits

C. Step-down voltage regulators

D. Audio amplifiers

10. When the switch in a boost converter is turned OFF, the inductor:

A. Stops conducting current

B. Releases stored energy to the load through the diode

C. Short-circuits the supply

D. Reverses voltage permanently

11. What is the main function of a buck-boost converter?

A. Only step up DC voltage

B. Only step down DC voltage

C. Either step up or step down DC voltage

D. Convert AC to DC

12. In a basic (inverting) buck-boost converter, the output voltage polarity is:

A. Same as input voltage

B. Opposite to input voltage

C. Alternating

D. Zero

13. In a buck-boost converter, energy is stored mainly in the:

A. Capacitor

B. Diode

C. Inductor

D. Switch

14. The output voltage of an ideal buck-boost converter depends mainly on:

- A. Load resistance
- B. Input voltage only
- C. Duty cycle of the switch**
- D. Switching frequency only

15. Which of the following is a common application of a buck-boost converter?

- A. Voltage regulation in battery-powered devices**
- B. High-voltage AC transmission
- C. Audio signal processing
- D. RF communication systems

16. What is the primary function of a UPS?

- A. Increase the input voltage
- B. Provide uninterrupted power during outages**
- C. Convert AC to DC permanently
- D. Reduce power consumption

17. Which component in a UPS stores electrical energy for backup power?

- A. Transformer
- B. Rectifier
- C. Battery**
- D. Inverter

18. During a power failure, a UPS supplies power to the load using:

- A. Rectifier
- B. Bypass switch
- C. Inverter**
- D. Surge protector

19. Which type of UPS provides zero transfer time between mains failure and backup supply?

- A. Offline UPS
- B. Line-interactive UPS
- C. Online UPS**
- D. Standby UPS

20. Which of the following is a common application of a UPS?

- A. Electric motor speed control
- B. Data centres and computer systems**
- C. Street lighting
- D. Radio broadcasting

21. What is the primary source of energy in a battery electric vehicle (BEV)?

- A. Internal combustion engine
- B. Hydrogen fuel cell
- C. Rechargeable battery pack**
- D. Alternator

22. Which component converts electrical energy from the battery into mechanical energy to drive the wheels?

- A. DC–DC converter
- B. Electric motor**
- C. Inverter
- D. Battery Management System

23. What is the main function of the inverter in an electric vehicle?

- A. Convert AC to DC
- B. Convert DC to AC**
- C. Step up battery voltage
- D. Store electrical energy

24. Regenerative braking in EVs helps to:

- A. Increase vehicle speed
- B. Reduce battery life
- C. Recover energy during braking**
- D. Replace mechanical brakes completely

25. Which of the following is a key advantage of electric vehicles?

- A. Higher fuel consumption
- B. Zero tailpipe emissions**
- C. More engine noise
- D. Frequent oil changes

26. In the block diagram of an electric vehicle, which component stores electrical energy?

- A. Electric motor
- B. Power controller
- C. Battery pack**
- D. Inverter

27. Which block in an EV converts DC power from the battery into AC power for the motor?

- A. DC–DC converter
- B. On-board charger

C. Inverter

D. Battery Management System

28. What is the main function of the DC–DC converter in an EV block diagram?

A. Convert AC to DC

B. Convert DC to AC

C. Step down high-voltage DC to low-voltage DC

D. Store electrical energy

29. In the EV block diagram, regenerative braking sends recovered energy back to the:

A. Electric motor

B. Battery pack

C. Inverter only

D. Charging port

30. Which block manages battery safety, state of charge, and temperature in an EV?

A. Motor controller

B. Inverter

C. Battery Management System (BMS)

D. Power distribution unit

31. Which component is the main source of electrical energy in an electric vehicle?

A. Fuel cell

B. Alternator

C. Battery pack

D. Generator

32. What is the primary function of the electric motor in an EV?

A. Store electrical energy

B. Convert mechanical energy to electrical energy

C. Convert electrical energy to mechanical energy

D. Control battery charging

33. Which EV component controls power flow between the battery and the motor?

A. Battery Management System

B. Power electronics controller

C. On-board charger

D. DC–DC converter

34. Which component is responsible for monitoring battery voltage, current, and temperature?

- A. Inverter
- B. Motor
- C. Battery Management System (BMS)**
- D. Charging port

35. The DC–DC converter in an EV is mainly used to:

- A. Convert DC to AC
- B. Step down high-voltage DC to low-voltage DC**
- C. Step up low-voltage DC to high-voltage DC
- D. Charge the traction battery

36. Which type of motor is most commonly used in modern electric vehicles?

- A. DC series motor
- B. Induction motor
- C. Permanent Magnet Synchronous Motor (PMSM)**
- D. Universal motor

37. Which motor is known for high efficiency and high power density in EV applications?

- A. Stepper motor
- B. Switched Reluctance Motor
- C. PMSM**
- D. DC shunt motor

38. Which motor does NOT require permanent magnets?

- A. PMSM
- B. BLDC motor
- C. Induction motor**
- D. PMDC motor

39. Which EV motor is well suited for regenerative braking?

- A. DC motor only
- B. AC induction motor
- C. PMSM
- D. Both B and C**

40. Which motor is preferred in some EVs due to its simple construction and robustness?

- A. Switched Reluctance Motor**
- B. Universal motor
- C. Synchronous motor
- D. Servo motor

UNIT-5

1. Which quantity in a magnetic circuit is analogous to electric current in an electric circuit?

- A. Magnetic flux density
- B. Magnetic field intensity
- C. Magnetic flux**
- D. Magnetomotive force

2. The opposition offered by a magnetic circuit to the establishment of magnetic flux is called:

- A. Reluctance**
- B. Reactance
- C. Resistance
- D. Permeability

3. Which of the following materials has the highest magnetic permeability?

- A. Air
- B. Cast iron
- C. Silicon steel**
- D. Copper

4. The dot convention in coupled coils is used to indicate:

- A. The physical location of windings
- B. The polarity of induced voltage**
- C. The direction of magnetic flux
- D. The magnitude of mutual inductance

5. If current enters the dotted terminal of the primary coil, the induced voltage in the secondary coil will be:

- A. Positive at the dotted terminal**
- B. Negative at the dotted terminal
- C. Zero at the dotted terminal
- D. Independent of dot marking

6. Two coils have dots on opposite ends. If currents enter both dotted terminals simultaneously, the mutual inductance will be:

- A. Zero
- B. Maximum and aiding
- C. Opposing**
- D. Independent of current direction

7. The working principle of a transformer is based on:

- A. Self-induction
- B. Mutual induction**
- C. Electromagnetic induction
- D. Electrostatic induction

8. A transformer that operates at a constant voltage and current rating is known as:

- A. Power transformer
- B. Distribution transformer**
- C. Instrument transformer
- D. Auto-transformer

9. Which type of transformer is used to measure high AC currents?

- A. Potential transformer
- B. Power transformer
- C. Current transformer**
- D. Isolation transformer

10. A step-up transformer is commonly used in:

- A. Domestic appliances
- B. Transmission of electrical power**
- C. Electric traction
- D. Distribution substations

11. The principle of operation of a DC motor is based on:

- A. Faraday's law of electromagnetic induction
- B. Fleming's right-hand rule
- C. Fleming's left-hand rule**
- D. Lenz's law

12. Which type of DC motor has high starting torque and is suitable for cranes and hoists?

- A. Shunt-wound DC motor
- B. Series-wound DC motor**
- C. Compound-wound DC motor
- D. Permanent magnet DC motor

13. In a shunt DC motor, the speed is:

- A. Highly variable with load
- B. Nearly constant regardless of load**
- C. Zero at no load
- D. Infinite at full load

14. What is the main function of a commutator in a DC motor?

- A. To increase torque
- B. To reverse current direction in the armature**
- C. To reduce armature resistance
- D. To control speed

15. A compound DC motor combines the characteristics of:

- A. Series and shunt motors**
- B. Series and synchronous motors
- C. Shunt and induction motors
- D. Stepper and synchronous motors

16. The working principle of a stepper motor is based on:

- A. Electromagnetic induction
- B. Magnetic attraction and repulsion of stator and rotor poles**
- C. Lorentz force
- D. Faraday's law

17. Stepper motors are commonly used when:

- A. High torque at high speed is required
- B. Precise position control is required**
- C. AC power is unavailable
- D. High efficiency is more important than accuracy

18. Which type of stepper motor moves in discrete steps without requiring feedback?

- A. Servo motor
- B. Hybrid stepper motor
- C. Permanent magnet stepper motor**
- D. Synchronous motor

19. Typical applications of stepper motors include:

- A. Electric vehicles
- B. CNC machines and 3D printers**
- C. Power transmission
- D. High-speed fans

20. The step angle of a stepper motor is defined as:

- A. Torque produced per step
- B. Angle rotated per input pulse**
- C. Total rotation in one revolution
- D. Rotor speed at full load

21. The primary purpose of a fuse in an electrical circuit is to:

- A. Regulate voltage
- B. Protect the circuit from overcurrent**
- C. Store electrical energy
- D. Convert AC to DC

22. A fuse operates on the principle of:

- A. Electromagnetic induction
- B. Melting of a low melting point conductor due to excessive current**
- C. Magnetic attraction
- D. Electrostatic force

23. Which type of fuse is used for protecting semiconductor devices like diodes and transistors?

- A. HRC (High Rupturing Capacity) fuse**
- B. Rewireable fuse
- C. Cartridge fuse
- D. Miniature fuse

24. What happens when the current exceeds the rated value of a fuse?

- A. The fuse cools down
- B. The fuse blows or melts, breaking the circuit**
- C. The fuse stores excess energy
- D. The fuse regulates the current

25. A rewirable fuse is preferred over a cartridge fuse because:

- A. It is more precise
- B. It is cheaper and can be reused after replacing the wire**
- C. It has higher breaking capacity
- D. It works for AC and DC circuits

26. The primary purpose of an MCB in an electrical circuit is to:

- A. Store electrical energy
- B. Protect the circuit from overcurrent and short circuits**
- C. Step up voltage
- D. Convert AC to DC

27. The two main protection mechanisms used in an MCB are:

- A. Thermal and magnetic**
- B. Mechanical and hydraulic

- C. Capacitive and inductive
- D. Optical and thermal

28. Which type of MCB trips instantly under high short-circuit current?

- A. Type B
- B. Type C**
- C. Type D
- D. All of the above

29. Compared to a fuse, an MCB has the advantage of:

- A. One-time use only
- B. Automatic resetting after a trip
- C. Can be manually reset without replacement**
- D. Slower response to overcurrent

30. In domestic wiring, the typical rated current for an MCB is:

- A. 1–5 A
- B. 6–32 A**
- C. 50–100 A
- D. 200–500 A

31. The main purpose of electrical earthing is to:

- A. Increase circuit voltage
- B. Protect human life and equipment from electric shocks**
- C. Improve power factor
- D. Reduce energy consumption

32. The point of a system which is connected to the earth is called:

- A. Neutral point**
- B. Live point
- C. Phase point
- D. Load point

33. Which type of earthing is commonly used in residential buildings?

- A. Plate earthing
- B. Rod earthing**
- C. Pipe earthing
- D. Strip earthing

34. Earth resistance should ideally be:

- A. High to prevent current flow
- B. Low to allow fault current to pass easily**
- C. Infinite
- D. Equal to load resistance

35. Which type of earthing uses a copper plate buried in the ground?

- A. Rod earthing
- B. Plate earthing**
- C. Strip earthing
- D. Pipe earthing

36. The difference between neutral and earth is:

- A. Neutral carries current in normal operation, earth carries current only in fault**
- B. Earth is always at higher potential than neutral
- C. Neutral is connected to metallic body, earth is not
- D. Neutral is optional in a circuit

37. Strip earthing consists of:

- A. Copper rod buried vertically
- B. Copper strip buried horizontally in the ground**
- C. Galvanized iron pipe
- D. Metallic plate in water

38. What is the main reason for connecting neutral to earth at the transformer?

- A. To stabilize voltage during fault**
- B. To increase current flow
- C. To reduce energy consumption
- D. To eliminate the need for MCBs

39. Which type of earthing is preferred in high-rise buildings?

- A. Rod earthing
- B. Plate earthing
- C. Strip earthing**
- D. Pipe earthing

40. When a fault occurs and the live wire touches the metallic body, the purpose of earthing is to:

- A. Reduce the current
- B. Provide a low-resistance path for fault current to flow to the ground**
- C. Increase voltage in the system
- D. Disconnect the neutral